

Investigation Into Improving PLONK Performance

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Abstract—Global visual geolocation is a critical tool for verifying media authenticity and recovering missing location metadata. This study investigates architectural improvements to PLONK, a diffusion-based geolocation model, by evaluating alternative vision encoders and spatial context integration. Using the OSV-5M dataset, the default StreetCLIP encoder was compared against DINOv2 and the highly compressed MobileCLIP2-S4. Additionally, multi-image fusion strategies—specifically averaging and attention-based combination of proximal image embeddings—were analyzed. The results demonstrate that while StreetCLIP achieved the lowest average distance error (1,076 km) due to its geographic pre-training, the lightweight MobileCLIP2-S4 model achieved remarkably comparable accuracy despite a significantly reduced parameter count. Furthermore, incorporating multiple proximal images failed to improve absolute localization accuracy; instead, the fusion process stripped fine-grained visual details and increased the model’s geographic uncertainty. While this multi-image approach successfully improved system-level coverage by distributing predictions across a broader, more representative geographic manifold, it ultimately worsened the average distance error. These findings indicate that single-image inputs paired with CLIP-derived encoders remain the optimal configuration for diffusion-based geolocation, and future multi-image strategies will require significantly larger clusters of proximal training data to effectively capture spatial context without spreading uncertainty.

I. INTRODUCTION

THE challenge of determining a precise location on Earth from a single image has gained mainstream recognition through platforms like GeoGuessr, highlighting the immense difficulty of global visual geolocation. Even human experts who dedicate years to the task struggle to achieve perfect accuracy. Beyond entertainment, however, the ability to accurately geolocate images has critical real-world applications. It is essential for recovering missing location metadata for historical archives, as well as for modern imagery where such data is simply unavailable. In practice, metadata is frequently lost during digital transmission, intentionally stripped by users seeking to preserve their privacy prior to distribution, or otherwise removed. Furthermore, with the rapid rise of generative AI, geolocation serves as a vital tool for journalists and investigators seeking to verify the

authenticity of visual media. By yielding a probabilistic distribution of potential geographic origins rather than relying on a brittle, single-point prediction, diffusion-based geolocation empowers these professionals to safely cross-reference plausible regions with other evidentiary sources.

While determining an image’s exact location is highly valuable, many images lack sufficient distinctive features to pinpoint a single coordinate with certainty. In these ambiguous scenarios, identifying a distribution of plausible locations, rather than forcing a single prediction, becomes enormously helpful.

This task is notoriously difficult due to the sheer vastness and visual diversity of the Earth. While localization is a highly active field in robotics, it typically focuses on contained, mapped environments where a robot uses sensors to navigate. Geolocation, by contrast, requires inferring a global position from pixels alone. To address the need for predicting multiple plausible locations, recent advancements have applied diffusion models to this space. Specifically, the PLONK architecture, from [1], treats geolocation not as a rigid classification problem, but as a diffusion process capable of returning diverse location predictions. This paper investigates potential improvements to the PLONK model by evaluating the impact of alternative vision encoders, such as DINOv2 [2] and MobileCLIP2-S4 [3], and exploring spatial context through multi-image fusion strategies.

II. OBJECTIVES AND SCOPE

A. Problem Definition

PLONK is a model which achieves image geolocation via diffusion [1]. The advantage of this approach is twofold. First, the model can return precise coordinate predictions rather than treating geolocation as a classification problem, as other models typically do. Second, diffusion allows the model to return multiple possible locations for a single image. For example, an image of a soccer field could correspond to any soccer field on Earth; diffusion allows the model to predict all of these plausible locations rather than committing to a single answer. This type of reasoning is much closer to how a person may solve this problem. The focus of this project was to determine whether there are improvements that can be made to the PLONK model.

I identified two primary axes along which PLONK could be improved: determining if the model’s performance could be improved by changing the image encoder, and evaluating whether providing the model with multiple images of the same location yields better predictions than a single image. Currently, StreetCLIP is the model used in the original PLONK paper for training on the OSV-5M dataset.

B. Assumptions and Scope

Several assumptions reduced the scope of this project. Outdoor scenes were the primary focus, as expanding to indoor environments significantly increases problem complexity. Consequently, the project was strictly limited to outdoor image geolocalization using widely available RGB street-view data. Indoor, satellite, and aerial imagery were entirely out of scope. Finally, regarding computational constraints, it was assumed all models could be efficiently trained and evaluated using a single, local NVIDIA GeForce RTX 4090.

C. Project Goals

The goals for this project were organized into three levels of increasing ambition, establishing concrete milestones and measurable outcomes. For **Level 1**, the experimental infrastructure was established, successfully reproducing the original PLONK baseline on the OSV-5M dataset [4], and swapping the default StreetCLIP encoder with alternatives to measure baseline performance changes. **Level 2** involved conducting a comprehensive comparison of alternative vision encoders (DINOv2, MobileCLIP2-S4) and evaluating multiple multi-image fusion strategies (averaging and attention-based combination). Finally, for **Level 3**, complex attention-based fusion mechanisms and architectural changes were evaluated to determine the optimal architectural configuration for improving overall geographic prediction accuracy.

III. METHODS

This section describes the specific algorithms, datasets, metrics, and experimental procedures used to pursue the goals outlined above.

A. Datasets and Proximity Grouping

I obtained an implementation of PLONK from [1] which was modified and trained for these experiments. For training, I used the OSV-5M dataset [4]. To evaluate the model, I utilized the train, validation, and testing splits established by the original PLONK authors.

The proximity grouping pipeline operated in two phases. In the first phase, a spatial index was constructed offline by extracting the GPS coordinates of every image in the dataset and projecting each coordinate onto a 3D unit sphere. A KD-tree was built over these Cartesian points, then queried at a chord-length threshold corresponding to a user-specified surface radius (e.g., 100 m). This yielded the set of all other images whose capture location fell within that radius for each image. This neighbor mapping, together with the full matrix of pre-computed image embeddings, was persisted to disk.

In the second phase, during training, each image was treated as an anchor and its pre-computed neighbor list was retrieved from the index. A random number of neighbors (between a configured minimum and maximum) were sampled, and their embeddings were fetched directly from the on-disk embedding matrix by integer index. To perform this fusion, the model architecture stacks vision encodings for previous observations. The anchor and neighbor embeddings were then fused—either by computing a weighted average into a single conditioning vector, or by assembling them so that a downstream attention pooler could perform the fusion. The result is that every training sample carried spatial context from nearby images without requiring those images to be co-located in the same data shard or loaded together at read time.

B. Architecture and Training

The model utilized for all of these tests adopts the framework proposed by [1], illustrated in Figure 1. I used the Riemannian Flow Matching architecture because, while flow matching models are extensions of diffusion models, they work by predicting the flow of the noise rather than the denoising step. This gets applied to the noise similarly to how it would be modeled in a vector field. Flow matching is more versatile than standard diffusion, as it allows the usage of any distribution for the randomness, whereas traditional diffusion models are limited to Gaussian distributions. Riemannian flow matching is an extension designed to work on any manifold of an object. This is ideal for modeling on Earth, as image geolocalization is fundamentally attempting to locate a position on the surface of the Earth (a 2D problem), rather than within the entire volume of the Earth (a 3D problem).

The core network consists of 12 layers with a hidden dimension of 512. While the base diffusion architecture remained unmodified, the primary experimental variable was the image encoder, which served as the conditioning input, c . Consequently, the conditioning input size

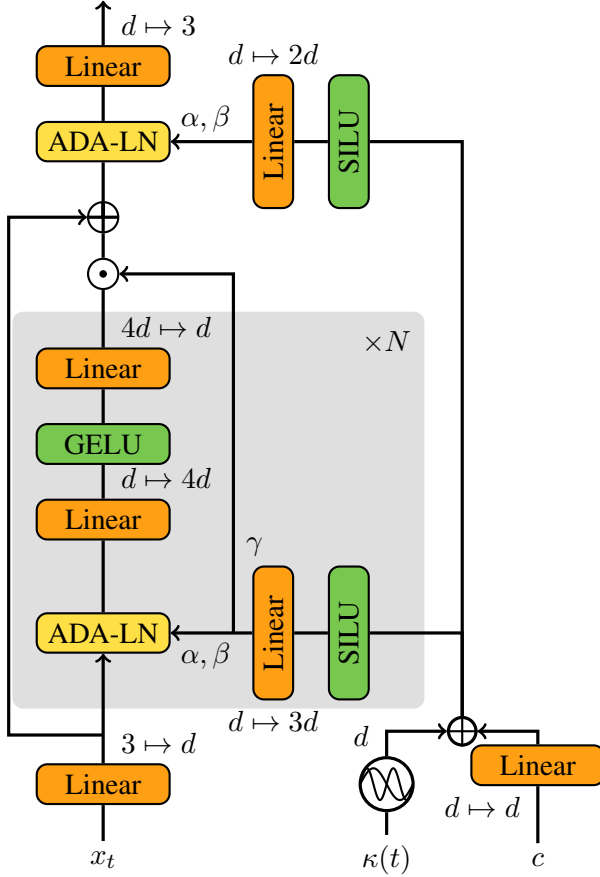


Fig. 1. Diffusion model architecture utilized in this paper. The core network design is reproduced directly from [1], as the base architecture remains unmodified for our experiments. The image encoder (providing conditioning input c) serves as our primary experimental variable.

was strictly dependent on the specific image encoder employed. A 10% dropout rate was applied to this image conditioning. Complete configuration files are available in the accompanying code repository.

The system was constructed on a spherical manifold using Riemannian Flow Matching for both the interpolant and loss, with a Classifier-Free Guidance (CFG) scale of 2. Optimization was performed using a Lamb optimizer ($LR=0.0008$, $\beta = (0.9, 0.999)$, universal weight decay=0.05) governed by a Warmup Cosine Decay scheduler across 1,000,000 training steps (500-step warmup). Training maintained an Exponential Moving Average (EMA) decay of 0.999 from the first step, and both training and inference phases employed a Sigmoid noise scheduler (ranging from -7 to 3, $\tau = 1$).

C. Evaluation Metrics

The primary metrics used to measure and compare performance across all experiments were the accuracy

by distances and areas (city, region, and country). I also measured the mean Haversine distance away from the true value. The Haversine distance is the shortest distance between two points on a sphere. Any mentions of mean distance error refer to this metric. These metrics are standard in the geolocalization literature and provide a clear, comparable measure of how close the model's predictions are to the true locations.

In addition to accuracy and average distance error, the precision, recall, density, and coverage of the model's predictions were measured. These metrics determine how well the predicted locations match the distribution of the actual locations. Precision measures the proportion of locations predicted by the model that fall within the manifold of the true locations. Recall measures the proportion of true locations that fall within the manifold of the predicted locations. Precision and recall are common metrics used in machine learning; however, because they typically refer to classification tasks, they must be modified for continuous applications.

In this case, precision and recall were measured using the manifolds of the object. Equation 1 shows the formulation for the manifold used to determine all retrieval metrics, where $B(z, Z)$ is the ball of the k -th nearest neighbor from that point in the dataset. Throughout this paper, the value of k was set to 3. Density represents the amount of predicted locations that fall within the manifold of the true locations. Finally, coverage is an improved recall metric specifically acting as a system-level recall for this pipeline.

$$\text{manifold}(Z) := \bigcup_{z \in Z} B(z, Z). \quad (1)$$

This manifold formulation and the calculations for these metrics were derived directly from [1].

D. Image Encoder Experiments

The first line of investigation concerned the image encoder. PLONK uses StreetCLIP as its default encoder; I systematically replaced this with alternatives to measure the effect on performance. The tested encoders included StreetCLIP [5], DINOv2 [2], and MobileCLIP2-S4 [3]. I initially trained with the StreetCLIP encodings to establish the baseline originally used by the paper. Next, the encoder was changed to DINOv2 to verify the pipeline worked end-to-end. I then trained and evaluated the model with all listed encoders and compared the results.

E. Multi-Image Fusion Strategies

The second line of investigation evaluated whether providing the model with multiple images from nearby

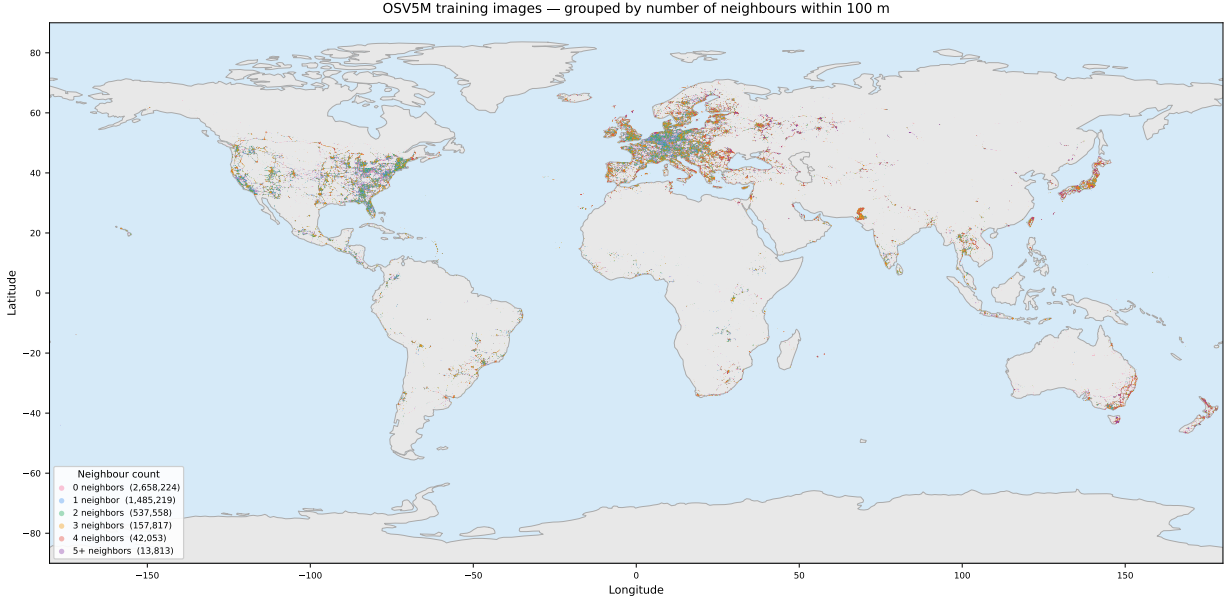


Fig. 2. Spatial distribution of the OSV-5M training dataset. Points are colored by the local density of neighboring images within a 100,m radius. The clustering highlights a significant geographic bias toward urban centers, major transit corridors, and regions within the USA and Europe.

locations improved predictions. Haas et al. [6] previously demonstrated that averaging four different image encodings could be effective for this purpose. I explored three strategies for combining images: averaging the encodings, utilizing an attention-based combination of encodings, and randomly selecting one image encoding at each diffusion step to determine if diversity in the conditioning signal was beneficial.

To implement the attention-based combination, a transformer with a single head was used. The specific image to be localized served as the query vector, while all of the neighboring image embeddings from the stacked vision encodings served as the keys and values. This allowed the transformer to intelligently combine the vectors using the target image’s context to dynamically weight the surrounding observations.

IV. RESULTS AND DISCUSSION

A. Summary of Objectives Achieved

This project successfully achieved the defined Level 1 and Level 2 objectives. For Level 1, the experimental infrastructure was fully established, and the baseline results of the original PLONK model using the StreetCLIP encoder on the OSV-5M dataset were successfully reproduced (as shown in Table III). For Level 2, alternative vision encoders (DINOv2 and MobileCLIP2-S4) were integrated and compared against the baseline to measure their effect on the average distance error. Furthermore, multi-image fusion strategies—including

averaging and attention-based mechanisms—were successfully implemented and evaluated to determine their impact on predictive accuracy.

B. Dataset Bias and Project Limitations

TABLE I
DISTRIBUTION OF TOP 5 COUNTRIES BY NEIGHBOR COUNT AND RELATIVE DATA FREQUENCY. NEIGHBORS ARE COUNTED BY IMAGES WITHIN A 100M RADIUS.

Group	#1	#2	#3	#4	#5
0 neigh. (54%)	USA 28%	Germany 7.4%	France 5.6%	Japan 3.5%	Russia 3.1%
1 neigh. (30%)	USA 24%	France 4.5%	Russia 4.4%	Japan 4.0%	Germany 3.9%
2 neigh. (11%)	USA 16.6%	Russia 7.4%	Japan 4.9%	Aus. 4.5%	France 3.4%
3 neigh. (3.2%)	Russia 12.9%	USA 9.3%	Aus. 6.7%	Japan 6.1%	Sweden 2.4%
4 neigh. (0.86%)	Russia 20.6%	Aus. 9.6%	Japan 7.6%	NZ 3.9%	USA 3.5%
5+ neigh. (0.28%)	Russia 35%	Aus. 13%	Japan 7.5%	NZ 6.8%	Finland 2.7%

Figure 2 illustrates the spatial distribution of the training dataset. Given the size of the dataset, it is representative of the entire combined dataset, and both validation and testing splits either match or are subsets of this distribution. Figure 2 shows a significant bias, with

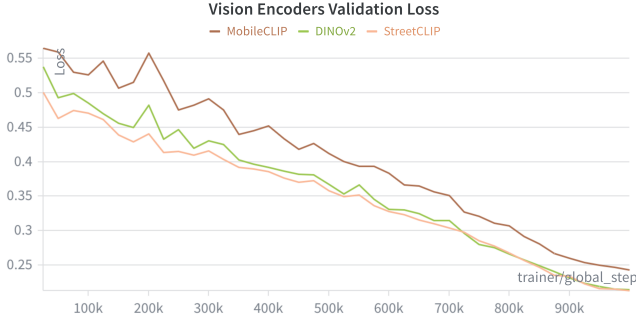


Fig. 3. Comparison of Vision Encoders Validation Loss by number of steps taken during training. Validation loss was measured every 25,000 steps.

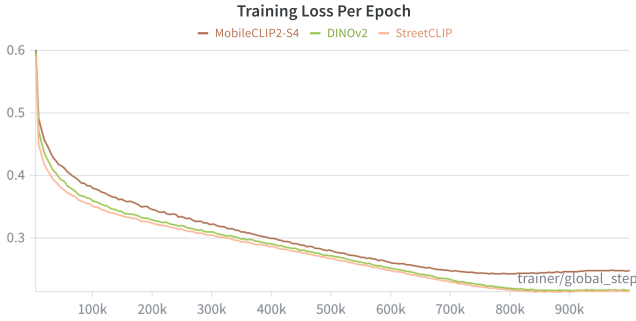


Fig. 4. Shows each of the encoders training loss per epoch over time by global step size.

large regions around the world lacking imagery data, juxtaposed against regions with highly concentrated image data. Specifically, the USA and Europe make up 24.70% and 35.46% of the data respectively (for a combined 60.16% of the entire training dataset), demonstrating a heavy bias toward the Western world. This highlights a major limitation of this project: there are significant geographic regions missing from the training data, and the model is highly likely to fail on images originating outside this distribution.

Table I reveals how the distribution of data changes by the number of neighbors an image has. While the USA dominates isolated image samples (0 neighbors), the distribution shifts heavily toward Russia for highly clustered data (3+ neighbors), with the USA dropping out of the top five entirely for 5+ neighbors.

These biases and constraints in the dataset directly limited the model’s generalizability and influenced the performance of the multi-image fusion experiments discussed in the following subsections.

TABLE II
OSV-5M TEST DATASET – DISTANCE METRICS

	geos. ↑	dist ↓	accuracy ↑ (in %, by radius)				
	/5000	(km)	1km	25km	200km	750km	2500km
Paper (StreetCLIP)	3767	1069	—	—	—	—	—
StreetCLIP	3783	1076	4.0	11.5	56.2	77.6	90.0
DINOv2	3648	1330	0.01	5.8	52.9	75.8	87.1
MobileCLIP2-S4	3481	1597	6.1×10^{-5}	4.7	49.6	71.5	84.3

TABLE III
OSV-5M TEST DATASET – ADMINISTRATIVE ACCURACY

	accuracy ↑ (in %, by admin.)			
	country	sub-region	region	city
Paper (StreetCLIP)	76.2	N/AN	44.2	5.4
StreetCLIP	76.5	9.9	44.1	5.2
DINOv2	73.1	7.2	39.1	3.5
MobileCLIP2-S4	69.0	6.7	36.5	3.4

C. Encoder Comparison

Table III demonstrates that I was able to successfully recreate the baseline performance seen in the paper. While the model trained with the same StreetCLIP encoder varied slightly in its performance metrics from the original authors’ results, this variance is expected, as they chose data points stochastically, meaning their final weight values will naturally differ.

While DINOv2 suffered at smaller distances, the larger the distance, the closer in accuracy it was to StreetCLIP. Its average distance error was not far off from StreetCLIP, with a difference of less than 300 km. This makes sense, as StreetCLIP was explicitly trained to encode geographic location data alongside image information, giving it a large edge. Another reason for the difference lies in the training methodology. While all encoders used in this paper rely on a Vision Transformer architecture, CLIP-derived models (like StreetCLIP and MobileCLIP2-S4) are designed to align with text inputs. DINOv2, conversely, is trained to reconstruct missing parts of an image. This means StreetCLIP and MobileCLIP2-S4 are likely better at encoding textual information (like street signs) already present in the image. While DINOv2 excels at spatial understanding, it lacks the specific symbolic encoding of CLIP models, which is likely why it suffers at shorter radiuses but captures comparable geographic context at broader distances.

MobileCLIP2-S4 performed surprisingly well. For a model specifically designed to run on a smartphone with a reduced parameter count and embedding size, it was highly comparable to DINOv2. The largest difference in accuracy between the two across all metrics was only

TABLE IV
RETRIEVAL METRICS COMPARISON

Model	Precision	Recall	Coverage	Density
StreetCLIP	0.8358	0.9040	0.5942	0.7929
DINOv2	0.8307	0.9105	0.5798	0.7802
MobileCLIP2-S4	0.8212	0.9137	0.5669	0.7526

4.3% for distances under 750 km. For city-level accuracy, it only differed from DINOv2 by 0.1%. Given that the model has an embedding size of 768 (compared to 1024 for DINOv2 and StreetCLIP), it achieved near-parity while using only 75% of the embedding capacity.

The retrieval metrics (Table IV) also paint an interesting picture. All models demonstrated very similar performance across all metrics, with the largest difference being only 0.0273 between MobileCLIP2-S4 and StreetCLIP. This demonstrates that all models were similarly adept at matching the distribution of the test dataset.

Another interesting observation is that Figure 4 shows the training loss stopped improving after 850k steps. However, Figure 3 shows that validation loss continued to improve even after the training loss plateaued. This is likely due to the learning rate scheduler reaching its lowest point during this period. While smaller step improvements are expected with a lower learning rate, the continued decrease in validation loss is notable.

D. Multiple Images vs Single Image

Table V shows that both averaging and attention-based methods of combining image embeddings led to a drop in accuracy and average distance error. The more images provided to the model, the worse it performed on these primary metrics. However, while accuracy and average distance error suffered, the precision, recall, and coverage metrics all improved with an increase in the number of images used. Specifically, coverage (acting as our system-level recall) significantly improved for both fusion methods. What this means is that the spatial distribution of the predictions better matched the ground truth manifold, even though the absolute center of those predictions was further away.

This indicates that combining images successfully allowed the model to better match the underlying geographic distribution of the test dataset. However, it also suggests that finer details were stripped away during the combination process. This means that while the model recognized the broad regional context better, the fine-grained details needed to pinpoint the exact coordinate were lost during fusion.

The most likely culprit for this result is the sheer lack of training samples containing many neighbors. Given that the models performed near the baseline when only inputting one image embedding (for both attention and averaging), this indicates a direct correlation between the volume of training data and the amount of images used. The sample size in the testing dataset decreases by 85.2% when moving from one image to two images, directly contributing to worse performance.

Another likely culprit stems from the nature of the OSV-5M dataset. A large amount of the data comes from dash-cams, meaning that images taken in close proximity are highly likely to be facing the exact same direction. If the camera is facing the same direction on a highway, proximal images are unlikely to provide any novel information that the anchor image did not already provide.

E. Comparing Single vs Multiple Image Accuracy



Fig. 5. Proximity batch near Terney, Russia. Leftmost is the anchor; the remaining five are nearby neighbors fused into a single prediction.

Figure 7 demonstrates a specific instance where the averaged embeddings model actually predicted the image’s location better than the model relying solely on the anchor image. This proves that, at the very least, utilizing multiple images *can* improve predictions in certain contexts. However, it is also evident that the model’s overall certainty fell. It produced predictions across a much wider geographic spread. It appears the resulting effect of averaging the embeddings is that the model becomes less certain, distributing its guesses over a larger area. This spread of uncertainty is likely the primary reason the multi-image models suffered worse average performance compared to using fewer images; while a large cluster of guesses may be closer to the true point, the guesses placed at extreme distances heavily penalize the average distance error. This observation is backed by Table V, which shows that city-level accuracy does not actually decrease by much even when a greater number of neighbors are introduced.

The attention-based model demonstrates similarly dispersed predictions for the multi-image input (Figure 8). Interestingly, the attention-based predictions showed very little change between the single- and multi-image inputs. This indicates that the attention mechanism was unable to effectively leverage the additional contextual

TABLE V

PERFORMANCE COMPARISON OF STREETCLIP AVERAGING VS. ATTENTION COMBINED ACROSS k NEIGHBORS. RETRIEVAL METRICS REVEAL THE TRADE-OFF BETWEEN GEOGRAPHIC ACCURACY AND COVERAGE (OMITTED FOR $k = 5$ DUE TO INSUFFICIENT DATA).

	$k = 0$		$k = 1$		$k = 2$		$k = 3$		$k = 4$		$k = 5$	
Metric	Avg	Attn	Avg	Attn	Avg	Attn	Avg	Attn	Avg	Attn	Avg	Attn
Sample Size (N)	212,992		31,444		5,445		1,370		395		118	
<i>Categorical Accuracy (%)</i> $\uparrow$												
City	5.26	5.11	4.35	4.04	2.90	2.44	1.75	1.61	3.29	2.53	4.24	3.39
Region	43.83	43.55	35.94	34.85	26.50	26.28	19.49	18.54	16.20	14.94	12.71	20.34
Country	76.69	76.13	71.23	69.87	65.49	64.08	58.69	59.34	49.37	54.18	43.22	50.85
<i>Retrieval Metrics (%)</i> $\uparrow$												
Precision	83.76	84.14	92.36	92.64	94.90	94.46	96.58	96.30	96.75	96.25	–	–
Recall	91.06	90.72	87.89	87.87	92.55	92.09	95.04	95.06	97.75	97.75	–	–
Coverage	59.54	59.29	68.84	69.01	79.08	78.93	84.29	82.33	91.00	88.75	–	–
<i>Distance & Scoring</i>												
Haversine (km) $\downarrow$	1070.0	1096.2	1330.1	1376.8	1612.3	1660.1	1618.0	1813.6	1640.0	1893.3	1645.0	1717.6
GeoGuessr $\uparrow$	3775.9	3763.9	3451.3	3414.7	3089.5	3042.7	2863.6	2787.4	2726.8	2630.4	2683.7	2713.3

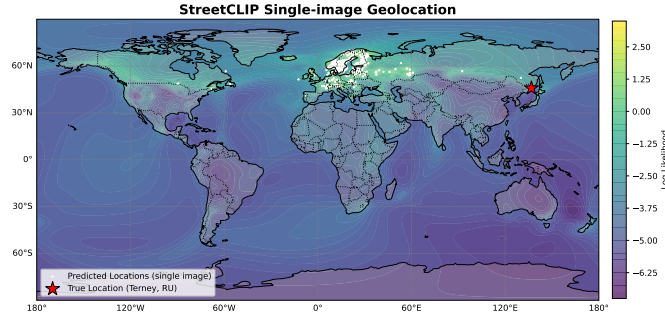


Fig. 6. Shows the log likelihood of the anchor image from Figure 5 being in a given location on Earth using the StreetCLIP model trained on individual images from Section III-E.

data to improve its estimates, instead retaining the problem of overall increased uncertainty. Furthermore, the attention-based model exhibited greater uncertainty on its single-image inputs compared to the baseline, demonstrating that the attention-based combination method suffered a higher performance penalty than the averaging-based method.

This increase in uncertainty is likely due to fine details being stripped away during the fusion process. Because the training dataset lacks data points with 5 or more neighboring images (making up only 0.28% of the training dataset), the model simply did not have the volume of data required to learn how to fuse them effectively. Furthermore, the statistical likelihood of the model being exposed to multiple images during training was inherently low. Because images were randomly selected during the training process, the model only had an approximate 25% chance of being exposed to more

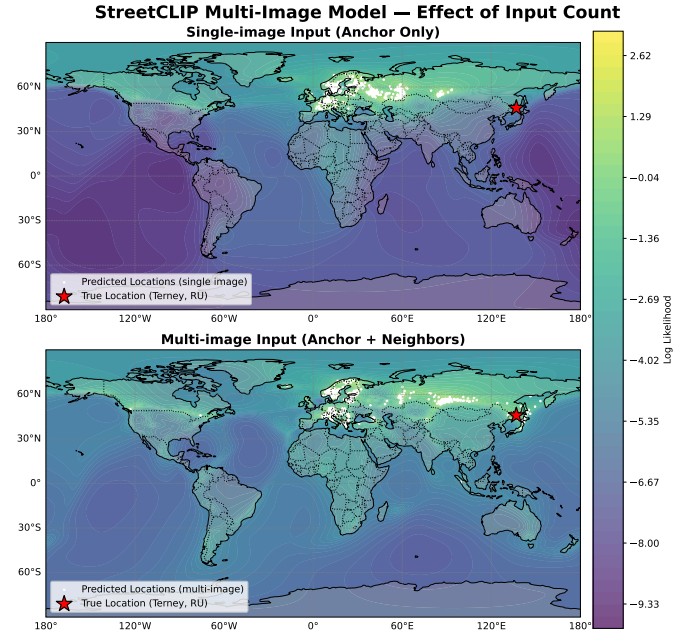


Fig. 7. Shows the log likelihood of the images being in a given spot on Earth using the StreetCLIP mean embedding model. The images come from Figure 5. The top one contains just the anchor image as input. The bottom one uses both the anchor and all neighboring images as inputs into the model.

than one image at any given instance. Given this massive discrepancy in the training data, it is extremely likely that the model over-optimized for individual images rather than combined embeddings. Future attempts to properly test multi-image embeddings will require training on datasets that feature a significantly higher concentration of proximal images.

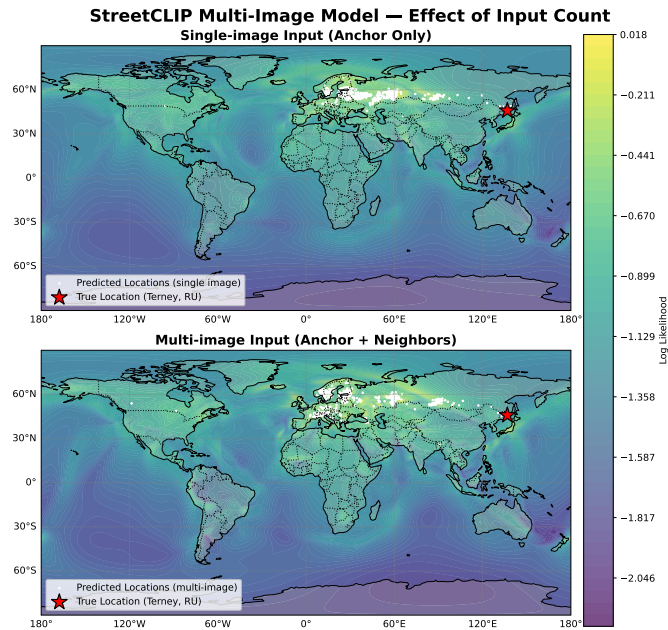


Fig. 8. Shows the log likelihood of the images being in a given spot on Earth using the StreetCLIP attention combination of embeddings model. The images come from Figure 5. The top one contains just the anchor image as input. The bottom one uses both the anchor and all neighboring images as inputs into the model.

V. CONCLUSION AND FUTURE WORK

This project successfully reproduced the PLONK architecture on the OSV-5M dataset, establishing a baseline for diffusion-based image geolocation. Comparing alternative vision encoders revealed that while StreetCLIP provided the lowest average distance error due to its explicit geographic pre-training, smaller models like MobileCLIP2-S4 achieved remarkably comparable accuracy with a significantly reduced parameter count. Conversely, multi-image fusion strategies—using both attention and averaged embeddings—failed to improve spatial accuracy. Instead, providing the model with multiple proximal images increased its geographic uncertainty, causing it to distribute predictions over a wider area and subsequently worsening the average distance error.

Future work will focus on fine-tuning the vision encoders during the diffusion training process, rather than relying on frozen, pre-trained weights. Fine-tuning spatial reconstruction models like DINOv2 could allow them to learn geolocation-specific features, potentially closing the performance gap with CLIP-based architectures. Additionally, evaluating these multi-image fusion strategies while simultaneously training the image encoder could yield better contextual blending. Finally, future experiments will utilize the substantially larger YFCC100M dataset to determine if a greater volume of geographically diverse training data mitigates the

uncertainty observed in the current multi-image proximity batches and improves overall prediction accuracy at smaller tolerance radii.

CODE AND DATASETS

The code and datasets are found [here](#).

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